

# The Wall Atlas: A curated record of barrier rejections encountered by an autonomous P-vs-NP proposer

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## Abstract

We curate the 32-record output of the four-barrier filter that gates every conjecture proposed by the **SperimentalMath** autonomous engine, deduplicate it to 20 unique conjectures, and tabulate which conjectures are blocked by which classical barrier. Of the 20 unique entries, three are textbook “calibration” attempts that the engine keeps re-emitting (and the filter keeps correctly catching) — diagonalization for **P** vs. **NP** [2], sum-check arithmetization for  $\mathbf{NEXP} \not\subseteq \mathbf{P}/\text{poly}$  [1], and the Karp-Lipton-style direct  $\mathbf{NEXP} \subseteq \mathbf{P}/\text{poly}$  assertion [4]. The remaining 17 entries are “genuine” proposer attempts that nonetheless trigger one of the four barriers; we group them by the underlying mathematical idea and identify the recurring failure modes. The contribution is curation, not a result; the corpus is offered as raw material for the design of better proposer prompts and better barrier-detection heuristics.

## 1 Introduction

The four classical barriers to circuit lower bounds and complexity-class separations — relativization [2], natural proofs [7], algebrization [1], and the Karp-Lipton-style implied collapse [4] — together rule out a large class of generic proof strategies. Most working complexity theorists eventually internalize the rules of thumb associated with these barriers: *don’t diagonalize black-box machines, don’t use only constructive-large-useful properties, don’t arithmetize sum-check into PSPACE-style protocols, don’t directly assert collapses that contradict standard hierarchy assumptions*. An autonomous proposer agent has no such internalization; it generates conjectures from natural-language prompts and has to be reminded, at each turn, by a barrier-detection module.

The Wall Atlas is the audit log of such reminders. Every time the proposer of **SperimentalMath** emits a conjecture, two LLM-driven barrier checkers (using independent backends) read the conjecture and its rationale, vote on each of the four barriers, and record a verdict  $\in \{\text{HITS}, \text{UNCERTAIN}, \text{SAFE}\}$  with a confidence score. A conjecture is rejected if either checker votes HITS on any barrier with confidence  $\geq 0.7$ .

This paper curates the raw rejection log into a deduplicated, classified catalogue. The intended audience is twofold:

- *Designers of autonomous research engines*: the patterns in §6 suggest concrete improvements to proposer prompts.
- *Complexity theorists curious about borderline cases*: some of the 17 genuine entries (§5) propose ideas that are not obviously blocked by the cited barrier; in those cases the barrier-checker may itself be wrong.

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\*The proposer agent and barrier-checker agents whose log we curate here are part of the autonomous engine behind **SperimentalMath**. Curation, deduplication, and the analysis of patterns are presented under the `MULTIAGENT_PIPELINE.md` framework as a survey-grade artefact, not a research result.

## 2 The four barriers, in one paragraph each

**Relativization (Baker–Gill–Solovay 1975 [2]).** A proof technique *relativizes* if its conclusion remains valid when every Turing machine in the argument is given access to the same oracle. Since there exist oracles  $A, B$  with  $\mathbf{P}^A = \mathbf{NP}^A$  and  $\mathbf{P}^B \neq \mathbf{NP}^B$ , no relativizing argument can resolve  $\mathbf{P}$  vs.  $\mathbf{NP}$ . Generic Turing-machine diagonalization is the canonical example.

**Natural proofs (Razborov–Rudich 1997 [7]).** A property  $P$  of boolean functions is *natural* (against  $\mathbf{P}/\text{poly}$ ) if it is *constructive* (poly-time computable from the truth table), *large* (holds for at least a  $1/\text{poly}$  fraction of functions), and *useful* (its truth on a function  $f$  implies that  $f$  has super-polynomial circuit size). Under standard cryptographic assumptions (one-way functions exist), no natural property is useful. Hence any property satisfying the three criteria above cannot be the basis of a circuit lower bound.

**Algebrization (Aaronson–Wigderson 2008 [1]).** A proof technique *algebrizes* if its conclusion remains valid when every boolean function in the argument is replaced by a polynomial extension (over a field of large enough size). Sum-check,  $\mathbf{IP} = \mathbf{PSPACE}$ -style arithmetization, and similar techniques algebrize. There exist two oracles  $A, \tilde{A}$ , with  $\tilde{A}$  a polynomial extension of  $A$ , that algebrize  $\mathbf{P}$  vs.  $\mathbf{NP}$  in opposite directions; hence algebrizing arguments cannot resolve  $\mathbf{P}$  vs.  $\mathbf{NP}$ ,  $\mathbf{NEXP}$  vs.  $\mathbf{P}/\text{poly}$ , or similar separations.

**Karp-Lipton-style implied collapse [4].** The Karp-Lipton theorem says that  $\mathbf{NP} \subseteq \mathbf{P}/\text{poly}$  implies the polynomial hierarchy collapses to  $\Sigma_2^P$ . By a chain of similar results, conjectures of the form “every  $\mathcal{C}$ -language has polynomial-size circuits” typically imply unlikely collapses in higher complexity classes, putting them in contradiction with standard separation assumptions.

## 3 Corpus and methodology

The raw barrier-checker log records 32 rejection events. Many events re-evaluate the *same* conjecture title, since the proposer re-emits a few canonical attempts every few hours; deduplication by conjecture title yields 20 unique conjectures (Table 1).

| Barrier        | Unique conjectures | Total rejection events |
|----------------|--------------------|------------------------|
| ALGEBRIZATION  | 7                  | 10                     |
| KARP_LIPTON    | 1                  | 3                      |
| NATURAL_PROOFS | 11                 | 12                     |
| RELATIVIZATION | 1                  | 7                      |
| <b>Total</b>   | 20                 | 32                     |

Table 1: Counts after curation. The two single-conjecture rows (KARP\_LIPTON and RELATIVIZATION) reflect calibration attempts; the natural-proofs and algebrization rows contain the genuine population.

## 4 Calibration entries

Three distinct conjectures account for roughly half (16/32) of the raw rejection events. Each is a textbook “don’t try this” attempt, which the proposer keeps re-emitting and which the barrier checker keeps correctly catching. We list them here so the rest of the catalogue can focus on genuine attempts.

1. **Diagonalization separation of  $\mathbf{P}$  and  $\mathbf{NP}$**  (5 events, RELATIVIZATION, max confidence 0.99). Proposer keeps trying “By Turing-machine diagonalization, there exists  $L \in \mathbf{NP} \setminus \mathbf{P}$ .” Caught by Baker–Gill–Solovay [2] as a black-box diagonalization argument.
2. **Every  $\mathbf{NEXP}$  language has polynomial-size circuits** (3 events, KARP-LIPTON, max confidence 0.99). Proposer keeps emitting the direct circuit-construction assertion; caught as a Karp-Lipton-style implied collapse.
3. **Sum-check bound for  $\mathbf{NEXP}$  lower bound** (4 events, ALGEBRIZATION, max confidence 0.95). Proposer keeps trying “use  $\mathbf{IP} = \mathbf{PSPACE}$ -style arithmetization on the  $\mathbf{NEXP}$  machine”; caught as algebrizing [1].
4. **Truth-table discriminant for circuit hardness** (4 events, NATURAL-PROOFS, max confidence 0.95). Proposer keeps proposing a poly-time-computable property of truth tables that holds on  $1/\text{poly}$  fraction of functions and implies super-polynomial circuit size — a textbook natural property [7].

These four account for 16 of the 32 raw events. **Design recommendation:** a single line in the proposer prompt — “do not re-emit any of the following four canonical traps” — would eliminate half the false starts.

## 5 Genuine attempts (the interesting half)

The remaining 16 unique conjectures are bona-fide attempts to construct a complexity-theoretic invariant from a fresh mathematical angle. We group them by the cited barrier and reproduce the proposer’s self-described *rationale* (the one-sentence motivation that accompanies each conjecture). Confidence scores are the combined two-LLM verdict; entries are sorted within each section by descending confidence.

### 5.1 Natural-proofs hits (10 entries)

Most of these propose a single-number invariant of the truth table (spectral, magnitude-theoretic, ergodic-theoretic, etc.) and conjecture that it lower-bounds a complexity measure. The natural-proofs flag fires when the invariant is *poly-time-from-the-truth-table*, *spread over a non-negligible fraction of all functions*, and *useful* against  $\mathbf{P}/\text{poly}$ .

| Title                                                                     | Conf | Rationale                                                                                                            |
|---------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------|
| Stern-Brocot Rank of Truth Tables Lifts Average-to-Worst MCSP             | 0.88 | Stern-Brocot continued-fraction depth of a truth-table-as-rational, lifted to MCSP via Levin-style amplification.    |
| Mahler Measure of Truth-Table Polynomial Lower-Bounds $\text{DNF}_{\min}$ | 0.85 | Logarithmic Mahler measure of the Lagrange-interpolating polynomial of $f$ as a hardness witness for DNF size.       |
| Plünnecke Doubling of Top-Fourier Spectrum Bounds $\mathbf{ACC}^0$ Size   | 0.85 | Doubling constants of the top- $k$ Fourier spectrum of $f$ via Plünnecke-Ruzsa as $\mathbf{ACC}^0$ size lower bound. |

|                                                                                         |      |                                                                                                                                                             |
|-----------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monotone Ultra-limit Convergence of $\kappa$ along majorization completion              | 0.78 | Internal stress test of the CG-KW framework's $\kappa$ along an ultra-filter limit; flagged by NP because $\kappa$ becomes a natural property in the limit. |
| Leinster Magnitude of Hamming-Embedded Communication Matrices $\mathbf{R}(\text{DISJ})$ | 0.78 | Leinster magnitude of the Hamming-Hamming product space as a discrepancy lower bound.                                                                       |
| Bivariate Mahler Measure of Sign Polynomial Lower-Bounds $\mathbf{R}(\text{DISJ})$      | 0.78 | Bivariate Mahler measure of the sign polynomial of $f$ as a randomized communication lower bound.                                                           |
| Lasserre Moment-Rank Gap of Accept Set Bounds $\mathbf{AC}^0$ PARITY Size               | 0.78 | Lasserre moment-rank gap of the accept-set indicator polynomial.                                                                                            |
| Schur-Horn Majorization Gap Lower-Bounds Sign-Rank Discrepancy                          | 0.70 | SVD vs. row-sum majorization gap on the $\pm 1$ truth-table matrix.                                                                                         |
| Median-Restriction Hardness Amplification Gap for DNF Truth Tables                      | 0.70 | Median over restrictions of $f$ , accumulated as a hardness gap.                                                                                            |
| Diameter-Multiplicity Power Law under Controlled Refinement                             | 0.70 | Internal stress test of the CG-KW framework's covering invariants.                                                                                          |

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Table 2: Genuine `NATURAL_PROOFS` hits.

**Pattern.** Eight of these ten entries propose a single number computed from the truth-table and conjecture it lower bounds circuit complexity. This is the canonical Razborov-Rudich template, and the barrier-checker's verdict is consistent with the literature. Two of them (the bottom two from the CG-KW framework) are flagged *despite* the framework's axiom A3 explicitly being an anti-natural-proofs claim — see [5] for the internal tension this raises.

## 5.2 Algebrization hits (6 entries)

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| Title | Conf | Rationale |
|-------|------|-----------|
|-------|------|-----------|

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|                                                                      |      |                                                                                                                                    |
|----------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------|
| Bounded Arithmetic Complexity of Algebraic Closure in Finite Fields  | 0.95 | Closure-algebra over $\mathbb{F}_q$ as proof-complexity invariant; flagged because the closure operation algebrizes.               |
| Tropical Circuit Weight Accumulation Bound                           | 0.90 | Tropical-arithmetic weight transform on circuit gates; flagged because tropical arithmetic still admits polynomial-extension lift. |
| Tropical Circuit Homotopy Stability under Weight Accumulation        | 0.90 | Companion to the above; same algebrizing transform.                                                                                |
| Kronecker Coefficients of Rectangular $\text{GL}_n$ -Representations | 0.90 | GCT-style approach via Kronecker coefficients; flagged because the GCT framework algebrizes by construction [6, 3].                |
| Lower-Bound Determinantal Complexity                                 |      |                                                                                                                                    |
| Orbit Closure Dimension vs. Boolean Circuit Size                     | 0.90 | Dimension of $\text{GL}_n$ -orbit closure of the function-as-tensor; same GCT-flavored algebrization concern.                      |
| Moment Matrix Eigenvalue Gap and SOS Degree for Max-CUT              | 0.85 | SOS degree from spectral gap of the moment matrix; algebrizes via Lasserre lift.                                                   |

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Table 3: Genuine ALGEBRIZATION hits.

**Pattern.** Algebrization concerns are triggered whenever the proposer reaches into commutative algebra (Mahler measure, Kronecker coefficients, orbit closures, SOS hierarchies). The two GCT-flavored entries are subtle: the GCT programme [6] is explicitly designed to evade the natural-proofs barrier, but algebrization remains an open concern [3]. A nuanced barrier-checker would distinguish “trivially algebrizes” from “algebrizes-but-via-techniques-actively-being-developed”; the current binary flag conflates the two.

## 6 Patterns and design recommendations

**Calibration noise.** 16/32 events are the four canonical traps of §4. A static disqualifier list in the proposer prompt would remove this noise.

**Truth-table-statistics monoculture.** 8 of the 10 genuine NATURAL\_PROOFS entries reduce to “a single number computed from the truth table.” This is the most heavily-mined region of the proposer’s hypothesis space and the barrier reliably catches it. A useful next step is for the proposer to learn that “truth-table single-number” is essentially exhausted, and to bias toward *relational* or *categorical* invariants instead.

**Algebrization is the proposer’s blind spot.** The proposer has no obvious filter for “does my construction lift to a polynomial extension”. The 6 genuine algebrization hits are the symptom; a syntactic check (“does the construction use only ring operations”) in the proposer prompt would help.

**Internal tensions.** Two CG-KW framework sub-conjectures hit the natural-proofs barrier despite the framework’s A3 axiom claiming anti-natural-proofs immunity. This is a contradiction the framework must address before its sub-conjectures can be tested coherently.

## 7 What to do with the genuine entries

For each entry in §5, the obvious next step is to re-evaluate the barrier verdict by hand, with full attention to the specific construction. In the case of the GCT-flavored entries (§5.2, last two rows), this is a research-level question; in the case of the truth-table single-number entries it is mostly bookkeeping. We do not perform that re-evaluation here.

The corpus is offered as raw material; both the barrier-checker verdicts and the proposer’s rationales are publicly logged in the `barriers/` subdirectory of the [SperimentalMath](#) repository.

## Reproducibility

The deduplicated catalogue is generated by `papers/wall_atlas/curate_walls.py` from the four `barriers/*.jsonl` files in the [SperimentalMath](#) repository. The output `wall_atlas_curated.json` is committed alongside this paper.

## References

- [1] S. Aaronson, A. Wigderson. *Algebrization: A new barrier in complexity theory*. ACM Trans. Comput. Theory, 2009 (STOC 2008).
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